
Start: HmmFree
ptr == NULL OR !HmmIsLivePayload ptr?
Yes
No
Return / Do Nothing
block = HMM_BLOCK_OF ptr
Mark block as free via HmmSetTags size, 1
Call HmmCoalesceAndInsert block
block == NULL OR block->is_free == 0?
Yes
No
Return
Identify right_neighbor block + block->size
right_neighbor < g_heap_end AND right_neighbor is free?
Yes
No
Remove right_neighbor from free list, add size to new_size
Check if block != g_heap_start
Yes
No
Inspect left neighbor via footer / header calculation
left_neighbor >= g_heap_start AND left_neighbor is free?
Yes
No
Remove left_neighbor from free list, update new_size & block pointer
Apply HmmSetTags block, new_size, 1
block + new_size == g_heap_end?
Yes
No
Shrink simulated program break: g_heap_end -= new_size
Insert block into head of free list via HmmFreeListInsert
Return
Return